Project improvement audit | 29 October 2026

Coca-Cola Inventory & Distribution | Evidence-based supplement to the initial project package. All calculations below were performed locally from the unchanged 50,000-row source CSV. This is not a substitute for MySQL, Databricks or Power BI execution evidence.

1. Gaps in the earlier package

Deliverable	Current gap / required correction
Faculty report	Initial report is 8 pages, not the approximately 30 pages specified. Expand only with evidence and meaningful commentary.
MySQL and DBeaver	SQL written but not run against a live MySQL instance; obtain actual 50,000-row import validation and saved outputs.
Databricks	Notebook/code prepared; Delta table creation and Spark execution not verified.
Power BI	Dashboard specifications are not a PBIX; build three measures, four visuals and slicers with checked totals.
Dataset provenance	Original public publisher or permitted provider has not been documented.
GitHub and presentation	Repository and individual demonstration require real publication and proof.
Faculty database requirement	Guideline names PostgreSQL, while chosen project implementation is MySQL. Request approval.

2. Further observed checks

Check	Verified value	Interpretation
Rows closing at/below supplied reorder point	1,451	Potential review alerts, not necessarily orders.
Rows closing below safety stock	0	Investigate definitions and timestamps.
Overstock flag AND closing at/below reorder	0	Concurrent flags/thresholds require interpretation; overstock definition unknown.
Forecast above actual	24,389	Row-level counts, no known forecast horizon.
Forecast below actual	24,445	Forecast bias cannot be inferred without period details.
Forecast exactly equal actual	1,166	Exact matches.
Weighted absolute percentage error	7.97%	Sum of absolute errors divided by sum of recorded demand.

3. Source-derived market comparison

Market	Demand units	Revenue USD	Overstock rate
North America	1,716,516	1,494,425.32	85.5%
India	1,578,718	1,371,793.59	85.5%
Asia Pacific	1,529,720	1,329,067.04	84.8%
Latin America	1,484,863	1,293,448.42	85.5%
Europe	1,385,673	1,212,345.39	84.9%
Southeast Asia	1,372,666	1,198,730.72	85.2%
Middle East & Africa	1,247,693	1,086,574.93	85.5%
Australia	1,156,238	1,005,345.64	84.8%

4. Critical interpretation rules

The 85.2% overstock-flagged share is NOT automatically 85.2% of physical inventory held in excess. The dataset lacks the documented algorithm behind Overstock_Flag. Reorder-point and safety-stock figures are pre-existing fields, not independently optimized policy recommendations. Sales and demand match for every record, so any unobserved lost sales cannot be measured here.

The 3,841 inventory-balance discrepancies are based on an assumed equation: opening plus production minus sales minus waste equals closing. Inter-warehouse transfers, returns, adjustment events or time-boundary mismatches could explain deviations. Avoid correcting balances without system definitions.

5. Recommended work sequence

Step 1: Verify dataset publisher/permission and define the row grain (warehouse-product-date-market or otherwise). **Step 2:** Confirm how overstock, stock-out, service level, turnover, demand and costs were calculated. **Step 3:** Import exact CSV to MySQL 8 using DBeaver, run the validation suite and export real results. **Step 4:** Execute the Python notebook end-to-end and preserve its output. **Step 5:** Execute Databricks Delta pipeline and save screenshots. **Step 6:** Build and reconcile Power BI with the source totals. **Step 7:** Fill report evidence sections, GitHub link and team contributions; then create a live demonstration script.

6. Faculty-aligned report architecture

Pages	Chapter / evidence
1-5	Cover, abstract, business problem, scope and limitations
6-8	Data source, complete dictionary, quality checks, workflow
9-11	MySQL design, import, live DBeaver query screenshots
12-15	Python methods, 3+ charts, substantive observations

Pages	Chapter / evidence
16-18	Databricks / Delta methods and actual Spark output
19-22	Power BI KPIs, 4+ visuals, filters, screenshots
23-26	Cross-tool reconciliation; findings and recommendations
27-30	Conclusion, contribution, references, appendix and evidence

No project section should imply a system was executed without screenshots or logs. The expansion above is an evidence roadmap, not a claim that those later stages are finished.